

Name: _____ Date: _____

Problem 1. *For each of the following sequences, find the first few terms and then find the limit of the sequence.*

a) $a_n = \left(-\frac{1}{2}\right)^n$

b) $b_n = \frac{2^n}{n^2}$

c) $c_n = \log_2(n)$

Problem 2. *Again, find the first few terms of the sequence and compute the limit. The only difference is this time the sequences will be given as implicit (or recursive) formulas, rather than explicit ones.*

a) $a_1 = 1; a_{n+1} = 3a_n - 1$

b) $a_1 = 0; a_{n+1} = 1 - 2a_n$

c) $a_1 = \frac{-1}{2}; a_{n+1} = \frac{-1}{2}a_n$

Problem 3. A *geometric sequence* is a special type of sequence like the one given in problem 2c, where to get from one term in the sequence to the next you multiply each term by some constant. The number we multiply by, we may call the **ratio** and we typically use the variable r . A geometric sequence can then be written as $a_1 = r; a_{n+1} = ra_n$. Alternatively, you can write it explicitly as $a_n = r^n$. In this problem, your goal is to explore what happens as you change the ratio r . For which values of r does the geometric sequence diverge, and for which values does it converge (and what does it converge to)? You can let r be any real number – positive/negative, rational/irrational, etc.

Problem 4. Compute each of the following limits. You can either make a table or draw the graph.

a) $\lim_{x \rightarrow \infty} \frac{1}{x}$

b) $\lim_{x \rightarrow 0^+} \frac{1}{x}$

c) $\lim_{x \rightarrow 0^-} \frac{1}{x}$

d) Based on your answers to parts b and c, does the two-sided limit $\lim_{x \rightarrow 0} \frac{1}{x}$ exist?

Problem 5. *Compute each of the following limits. You can either make a table or draw the graph.*

a) $\lim_{x \rightarrow \infty} \frac{1}{x^2}$

b) $\lim_{x \rightarrow 0^+} \frac{1}{x^2}$

c) $\lim_{x \rightarrow 0^-} \frac{1}{x^2}$

d) Based on your answers to parts b and c, does the two-sided limit $\lim_{x \rightarrow 0} \frac{1}{x^2}$ exist?

Problem 6. *For each part, I want you to sketch the graph of a function which has the desired property. The goal of this problem is to understand what limits tell us about the graphs of functions.*

a) $f(x)$ is defined at $x = 2$, but $\lim_{x \rightarrow 2} f(x)$ does not exist.

b) $f(x)$ is undefined at $x = 2$, but $\lim_{x \rightarrow 2} f(x)$ exists.

c) $f(x)$ is defined at $x = 2$, the limit $\lim_{x \rightarrow 2} f(x)$ exists, but $\lim_{x \rightarrow 2} f(x) \neq f(2)$.

d) $f(x)$ is undefined at $x = 2$, $\lim_{x \rightarrow 2^-} f(x) = \infty$, and $\lim_{x \rightarrow 2^+} f(x) = -\infty$.